

Session 3 — Materials, CMF & Productive Direction

From requirements to material decisions, finishes, production logic and evidence.

Robust Design Sprint

Where Session 3 fits

Session 2 defined what the product must achieve. Session 3 starts defining what it could be made of, how it feels, and how it may be produced.

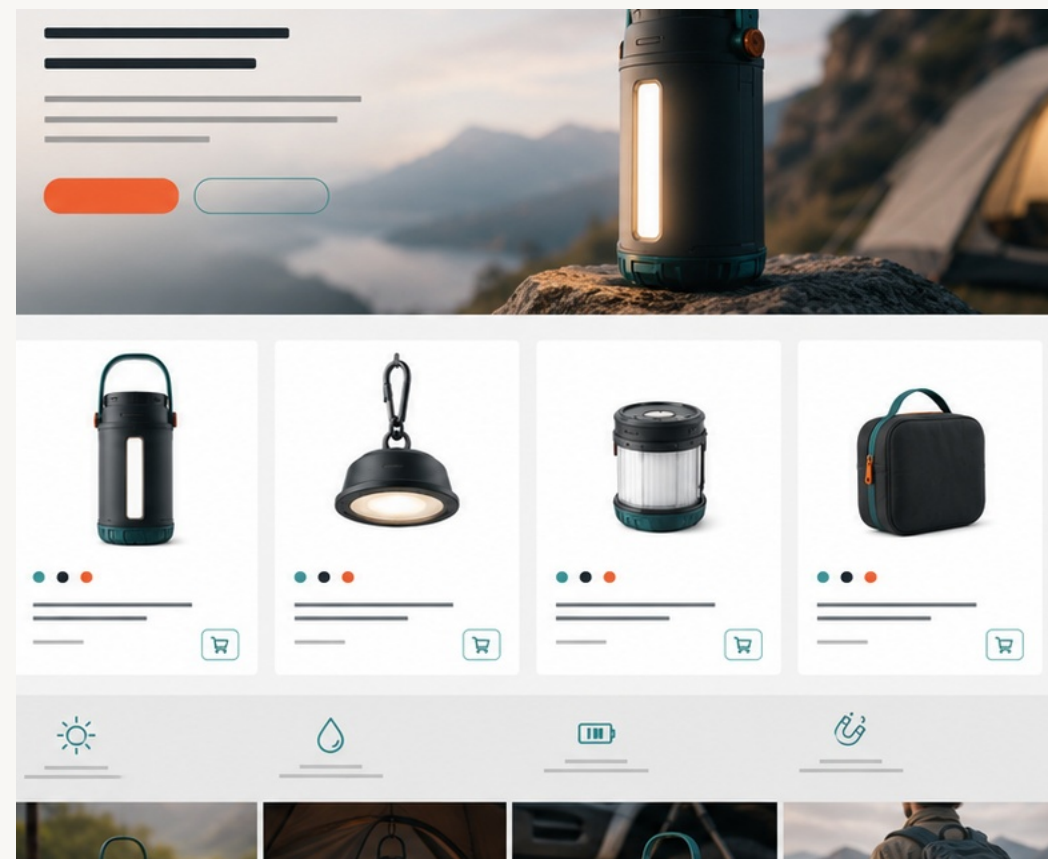


Today's goal

Same product, two languages.



Technical report: material logic for a manufacturer



Visual/commercial report: material story for the user

What we will define today

1. Material strategy

Which materials make sense for use, resistance, comfort, cleaning, cost and production?

2. CMF direction

What colors, materials and finishes support the product identity and user experience?

3. Production direction

Which processes could manufacture the main parts: sewing, injection, CNC, 3D printing, RF welding, etc.?

4. Material risks

What could fail: scratches, tears, leaks, weight, high cost, cleaning, water damage or weak joints?

5. Alternatives

What backup materials or processes could work if the first option is not viable?

6. Evidence needed

What photos, tests, references, samples, diagrams or comparisons will support the decision?

Deliverable: Materials, CMF and Productive Direction Report

How to choose materials

- **Use context**

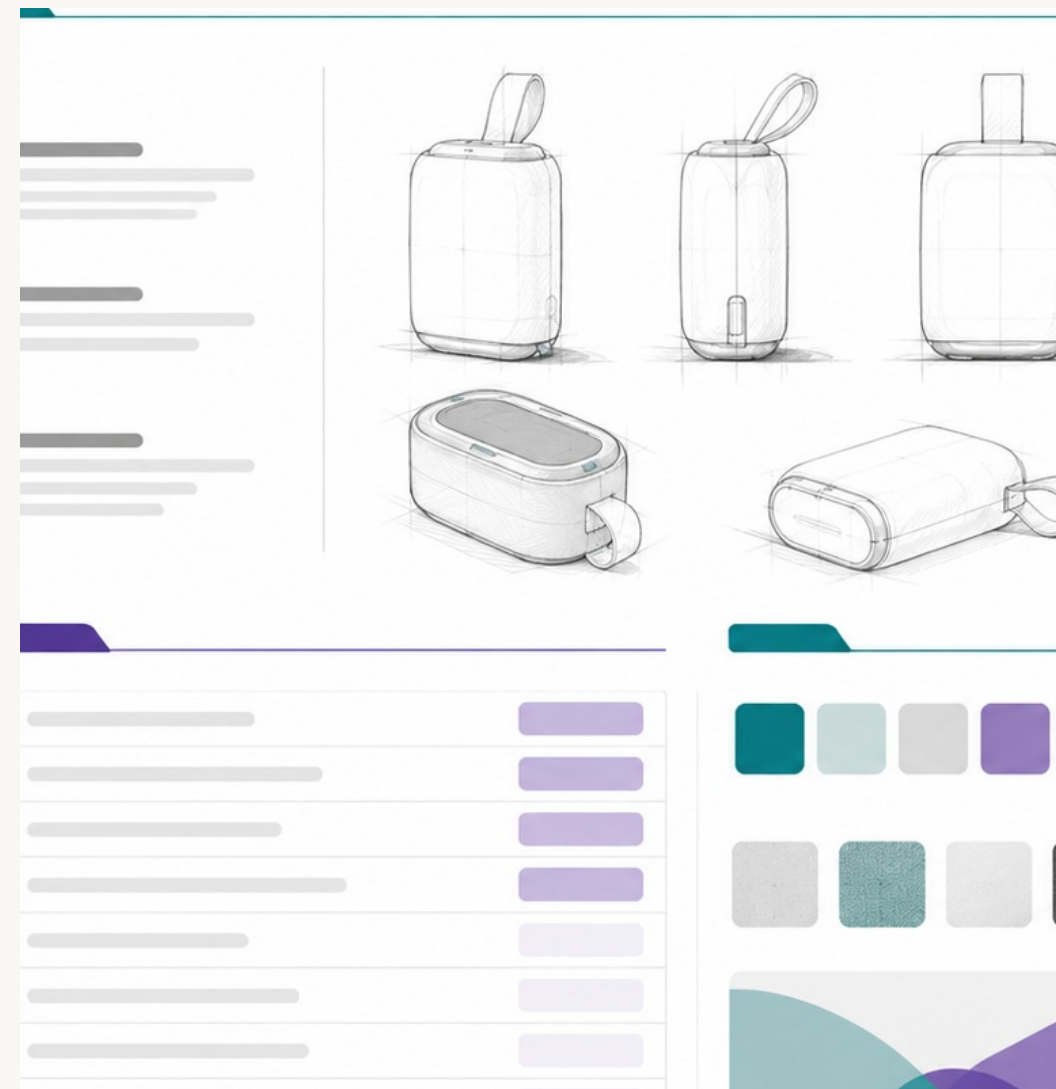
Water, movement, impact, friction, weight, cleaning, heat, outdoor use.
- **User body**

Grip, touch, comfort, pressure, sweat, posture, adjustability.
- **Production**

How it can be cut, sewn, molded, printed, welded, assembled.
- **Business**

Cost, suppliers, availability, repair, replacement parts.
- **Communication**

What the material says: durable, premium, light, safe, sustainable.



CMF is not decoration



CMF = Color + Material + Finish

It connects technical performance with product identity.

Color

Visibility, identity, safety, emotion, product family.

Material

Durability, comfort, weight, cleaning, sustainability.

Finish

Texture, grip, perceived quality, maintenance, scratches.

Example: “textured matte finish” can mean grip + durability + sport identity.

Session 3 deliverable checklist

Materials, CMF and Productive Direction Report

1. Cover / Header

Project name, team, course, session and main product image.

2. Material strategy

General explanation of the material needs of the product.

3. Materials table

Part, material, properties, process, finish, justification and risk.

4. CMF palette

Colors, materials, textures and finishes with short captions.

5. Material references

Photos, samples, benchmarks, suppliers or real material examples.

6. Production process direction

Possible processes for each major part or component.

7. Risks and alternatives

What could fail and what backup options exist.

8. Technical explanation

How this should appear in the report for a manufacturer.

9. Visual/commercial explanation

How materials will be shown to the final user.

10. Next steps

What to test, measure, draw, quote or confirm before Session 4.

Suggested length: 8–12 pages. Visual, concise, with tables, captions and clear evidence.

Main table for Session 3

This table turns your Session 2 requirements into material and production decisions.

Part / Component	Proposed Material	Key Properties	Process	Finish / Color	Risk / Alternative
Main body	to be defined	durable, light, safe	injection / sewing / RF welding	matte / textured	needs testing
Strap / support	to be defined	flexible, strong, comfortable	cutting + sewing	textile finish	check resistance
Connector / joint	to be defined	precise, repeatable, strong	injection / CNC / commercial	neutral / technical	check tolerance
User touch area	to be defined	soft, washable, ergonomic	foam / textile / rubber	grip texture	check cleaning
Replaceable module	to be defined	modular, easy to remove	assembly / fastener	brand color	check wear

Add images next to the table: material reference, component sketch, prototype photo or production example.

Do not claim “final material” unless you have evidence. Use “proposed material” or “to be tested”.

NotebookLM prompt for Session 3

Using the Session 2 Concept and Requirements report, create a material and CMF proposal for this product.

Include:

1. Material strategy
2. Main parts and possible materials
3. Material properties needed
4. Possible production processes
5. CMF palette direction
6. Material risks
7. Alternative materials
8. How to explain the materials in the technical report
9. How to communicate the materials in the visual/commercial report

Do not invent final specifications. If information is missing, write “to be defined” or “needs testing”.